

Name: _____

Date: _____

Solve each problem. Show your work in the box.

Write your final answer on the answer line (or in the blanks shown).

Use the strategies you learned in Lesson 5.12.03.

1 Read the Line Plot

A line plot shows siblings: 0 siblings = 2 marks; 1 sibling = 5 marks; 2 siblings = 4 marks; 3 siblings = 3 marks.

How many students were surveyed in total?

$2 + 5 + 4 + 3 = 14$ students. _____

2 Find the Mode from a Line Plot

Same line plot (0: 2 marks, 1: 5 marks, 2: 4 marks, 3: 3 marks).

What is the mode of the data?

1 sibling — it has the most marks (5). _____

3 Create a Line Plot

Heights of 10 plants (cm): 8, 10, 9, 10, 8, 11, 9, 10, 8, 11.

Create a line plot for this data. What is the most common height?

Marks at 8 (3 times), 9 (2 times), 10 (3 times), 11 (2 times). Most common: 8 cm and 10 cm (both appear 3 times — two modes).

4 Calculate the Mean from a Line Plot

Pages read: 5 pages (2 students), 6 pages (3 students), 7 pages (4 students), 8 pages (1 student).

Find the total pages read and the mean number of pages.

Total = $(5 \times 2) + (6 \times 3) + (7 \times 4) + (8 \times 1) = 10 + 18 + 28 + 8 = 64$. Mean = $64 \div 10 = 6.4$ pages.

5 Reasoning: Leo's Line Plot

Leo made a line plot of daily steps walked. He says the mode is the best measure to describe his typical day.

Do you agree? When might the mean be more useful than the mode?

The mode tells you the most frequent value, but the mean balances all values. If Leo had one very high day, the mode would